

## OpenIoT

OpenIoT is a scalable and flexible open source middleware for the Internet of Things. It enables the integration of various IoT devices and data sources, making it a versatile choice for IoT applications.

```
# Example: Using OpenIoT for data
ingestion
from openiot import connect
```

```
# Connect to OpenIoT platform
iot = connect('mqtt://broker.example.
com')
```

```
# Publish IoT data
iot.publish('iot/sensors/temperature',
payload='25°C')
```

## Eclipse IoT

Eclipse IoT is an ecosystem of open source projects that provide tools and frameworks for building IoT solutions. It includes projects like Eclipse Mosquitto (MQTT broker) and Eclipse Kura (IoT gateway).

```
// Example: Configuring Eclipse Mosquitto
as an MQTT broker
port 1883
listener 9001

# ...other configuration settings
```

## ThingsBoard

ThingsBoard is an open source IoT platform that offers device management, data visualisation, and rule-based automation. It simplifies IoT application development and provides a rich set of widgets for creating dashboards.

```
// Example: Creating a dashboard widget
in ThingsBoard
const temperatureWidget = new
TbWidgetV2(
  'temperature-widget',
  {
    title: 'Temperature',
    datasource: self.ctx.datasource,
    dataKey: 'temperature',
```

```
mainDataKey: 'temperature',
  }
);
```

## Popular open source IoT protocols

Efficient communication protocols are crucial in IoT, and open source has provided several robust options. Here are two widely used open source IoT protocols.

### MQTT (Message Queuing Telemetry Transport)

MQTT is a lightweight, open source protocol designed for efficient and reliable messaging in IoT. It follows a publish-subscribe model and is well-suited for constrained devices.

```
# Example: Using MQTT for IoT
communication
import paho.mqtt.client as mqtt

client = mqtt.Client()
client.connect("broker.example.com")

# Publish data
client.publish("iot/sensors/
temperature", "22.5°C")

# Subscribe to a topic
client.subscribe("iot/sensors/+data")
```

### CoAP (Constrained Application Protocol)

CoAP is an open source protocol designed for resource-constrained devices and low-power networks. It is similar to HTTP and is ideal for IoT applications that require a web-like interface.

```
// Example: Implementing CoAP server in
Java
CoapServer server = new CoapServer();
server.add(new
CoapResource("temperature") {
    @Override
    public void handleGET(CoapExchange
exchange) {
        // Respond with temperature data
```

```
exchange.respond("22.5°C");
    }
});
server.start();
```

## Examples of successful open source IoT projects

The open source community has birthed numerous IoT projects that exemplify the power of collaboration and innovation. Here, we'll dive into three successful open source IoT projects: Home Assistant, OpenHAB, and Node-RED. Let's explore how these projects leverage coding to create robust and user-friendly IoT solutions.

### Home Assistant

Home Assistant is a highly popular and versatile open source home automation platform that has gained a massive following in the IoT community. It's designed to simplify the management and automation of various smart devices in your home. Here's why Home Assistant stands out.

*Code example - Automation rule in Home Assistant (YAML)*

```
# Example: Turn on the lights when
motion is detected
automation:
  - alias: "Turn on lights when motion
detected"
    trigger:
      platform: state
      entity_id: binary_sensor.motion_
sensor
      to: "on"
    action:
      service: light.turn_on
      entity_id: light.living_room
```

In this YAML code, an automation rule is defined to turn on the lights when motion is detected by the motion sensor. Home Assistant allows users to create such rules to automate various aspects of their smart home.

### OpenHAB

OpenHAB is another prominent open